IMA Journal of Numerical Analysis (2011) **31**, 528−555
doi:10.1093/imanum/drp053
Advance Access publication on April 9, 2010

Stability analysis of block LDL^{T} factorization for symmetric indefinite matrices

HAW-REN FANG†
Department of Computer Science and Engineering, University of Minnesota, Minneapolis, MN 55455, USA

[Received on 5 September 2008; revised on 6 December 2009]

We consider block LDL^{T} factorization for symmetric indefinite matrices in the form LDL^{T}, where L is unit lower triangular and D is block diagonal with each diagonal block having dimension 1 or 2. The stability of this factorization and its application to solving symmetric indefinite linear systems has been well studied. On the other hand, while all rounding error analysis of block LDL^{T} factorization in the literature relies on the outer product form, this paper gives a novel componentwise backward error analysis based on the inner product form. The new results include a condition under which block LDL^{T} factorization in inexact arithmetic is guaranteed to preserve the inertia and a reliability analysis of rank estimation and inertia estimation of symmetric indefinite matrices by block LDL^{T} factorization.

Keywords: backward error analysis; numerical stability; block LDL^{T} factorization; symmetric indefinite matrix; inertia preservation; inertia estimation; rank estimation.

1. Introduction

A symmetric possibly indefinite matrix $A \in \mathbb{R}^{n\times n}$ can be factored into LDL^{T}, where L is unit lower triangular and D is block diagonal with each block of order 1 or 2. This is a generalization of the Cholesky factorization, which requires positive semidefiniteness. The process is described as follows. The case A being zero is trivial. The corresponding D is also zero and L is an identity matrix. When A is nonzero there exists a permutation matrix Π such that

$$\Pi A \Pi^{\mathrm{T}} = \begin{array}{c c} & \begin{array}{cc} s & n-s \end{array} \\ \begin{array}{c} s \\ n-s \end{array} & \begin{bmatrix} A_{11} & A_{21}^{\mathrm{T}} \\ A_{21} & A_{22} \end{bmatrix} \end{array},$$

where A_{11} is nonsingular, and $s = 1$ or 2 denotes that A_{11} is 1×1 or 2×2. The decomposition in outer product form is as follows:

$$\begin{bmatrix} A_{11} & A_{21}^{\mathrm{T}} \\ A_{21} & A_{22} \end{bmatrix} = \begin{bmatrix} I_s & 0 \\ A_{21}A_{11}^{-1} & I_{n-s} \end{bmatrix} \begin{bmatrix} A_{11} & 0 \\ 0 & \bar{A} \end{bmatrix} \begin{bmatrix} I_s & A_{11}^{-1}A_{21}^{\mathrm{T}} \\ 0 & I_{n-s} \end{bmatrix}, \tag{1.1}$$

where $\bar{A} = A_{22} - A_{21}A_{11}^{-1}A_{21}^{\mathrm{T}} \in \mathbb{R}^{(n-s)\times(n-s)}$ is the Schur complement. Iteratively applying (1.1) to the Schur complement and aggregating the results, we obtain a factorization in the form $PAP^{\mathrm{T}} = LDL^{\mathrm{T}}$, where P is a permutation matrix, L is unit lower triangular and D is block diagonal with each block of

†Email: hrfang@cs.umn.edu

© The author 2010. Published by Oxford University Press on behalf of the Institute of Mathematics and its Applications. All rights reserved.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

order 1 or 2. The overall decomposition requires $n^3/6$ multiplications and $n^3/6$ additions (subtractions), the same cost as the Cholesky factorization. In this paper we consider only the highest-order terms while counting the number of operations.

In block LDL^{T} factorization, choosing the permutation matrix Π and pivot size s at each step is called *diagonal pivoting*. As listed below, there are three types of pivoting methods for general symmetric matrices. See also Appendix A for a review.

1. Complete pivoting: the Bunch–Parlett pivoting (Bunch & Parlett, 1971).
2. Partial pivoting: the Bunch–Kaufman pivoting (Bunch & Kaufman, 1977) and a modification by Sorensen and Van Loan (Dongarra *et al.*, 1991, Section 5.3.2).
3. Rook pivoting: fast Bunch–Parlett and bounded Bunch–Kaufman (Duff & Reid, 1982, 1983; Ashcraft *et al.*, 1998).

Fang & O'Leary (2006) studied the block LDL^{T} factorization of a symmetric triadic matrix with above pivoting strategies, where a matrix is called *triadic* if it has at most two nonzero off-diagonal elements in each column. In addition, listed below are two pivoting strategies and their variants specifically for symmetric tridiagonal matrices. See also Appendix B for a review.

1. Bunch's method (Bunch, 1974, section 4) and a modification by Higham (1999, p. 183).
2. The Bunch–Marcia pivoting (Bunch & Marcia, 2005) and a simplified strategy (Bunch & Marcia, 2006).

Some historical remarks on the stability analysis of block LDL^{T} factorization are given as follows. Slapničar (1998, Section 7) derived a componentwise error bound for block LDL^{T} factorization with Bunch–Parlett pivoting. Higham (1997, 1999) analysed the stability of the Bunch–Kaufman strategy and Bunch's method. Bunch & Marcia (2005, 2006) showed the normwise backward stability of their pivoting strategy. Despite the rich literature, numerical properties of inertia preservation of block LDL^{T} factorization and its applications to rank estimation and inertia estimation are currently underexplored.

The inertia of a symmetric matrix is the triplet of integers consisting of the numbers of positive, negative and zero eigenvalues. By Sylvester's law of inertia (Golub & Van Loan, 1996, Theorem 8.1.17), the inertia is invariant under nonsingular congruence. More precisely, if two symmetric matrices A and B satisfy $A = CBC^{\mathrm{T}}$, where C is nonsingular, then A and B have the same inertia.

In exact arithmetic, block LDL^{T} factorization preserves the inertia of a given symmetric matrix A (i.e., the block diagonal matrix D has the same inertia as A). Note that, however, the Bunch–Kaufman pivoting (Bunch & Kaufman, 1977) does not guarantee the completion of block LDL^{T} factorization if the given matrix is rank deficient. When rounding errors are present the preservation of inertia is no longer guaranteed by any pivoting strategy. In particular, there could be a computed Schur complement not of full rank during the block LDL^{T} factorization, even if the given matrix is nonsingular.

The contribution of this paper is twofold. Firstly, we give a sufficient condition under which the block LDL^{T} factorization of a symmetric matrix is guaranteed to preserve the inertia in inexact arithmetic. It implies that, if a symmetric matrix is not too ill-conditioned, then its inertia is preserved in its computed block LDL^{T} factorization with the pivoting methods considered in this paper. Secondly, we analyse the stability of the block LDL^{T} factorization of a rank-deficient symmetric matrix. From this and from our experiments we suggest stopping criteria for rank estimation and inertia estimation in practice. Both stability analysis and experimental evidence indicate that the block LDL^{T} factorization, with chosen pivoting strategies, is reliable for rank estimation and inertia estimation. The last part can be regarded as a generalization of the results of Higham (1990) on rank estimation by the Cholesky decomposition with complete pivoting for symmetric semidefinite matrices.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

It is worth noting that all previous works on rounding error analysis of block LDL^{T} factorization rely on the outer product form. On the other hand, this paper gives a novel stability analysis based on the inner product form. By bridging the gap between the inner and outer product formulations in floating-point computation, our results remain valid for block LDL^{T} factorization in outer product form.

This paper is organized as follows. Section 2 gives the basics of rounding error analysis. Section 3 gives a backward error analysis, which leads to a condition under which block LDL^{T} factorization is guaranteed to preserve the inertia. The stability analysis to support the reliability of rank/inertia estimation by block LDL^{T} factorization for symmetric indefinite matrices is given in Section 4. Numerical experiments of rank/inertia estimation are reported in Section 5. The concluding remarks are listed in Section 6. Appendices A and B review the pivoting methods for general symmetric matrices and those for symmetric tridiagonal matrices, respectively.

Throughout the paper, without loss of generality, we assume that the required interchanges for any diagonal pivoting are done prior to the factorization, so that $A := PAP^{\mathrm{T}}$, where P is the permutation matrix for pivoting. A column vector of ones is denoted by e, and the 2×2 identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is denoted by I_2. We also denote the unit roundoff by u and use a 'hat' for each computed quantity. Inequality and absolute value for matrices are defined elementwise.

2. Basics of rounding error analysis

We use $fl(\cdot)$ to denote the computed value of a given expression and follow the standard model

$$fl(x \text{ op } y) = (x \text{ op } y)(1+\delta), \quad |\delta| \leqslant u, \quad \text{op} = +, -, \times, /, \tag{2.1}$$

where u is the unit roundoff. This model holds in most computers, including those using Institute of Electrical and Electronics Engineers standard arithmetic. Lemma 2.1 gives a basic tool for rounding error analysis (Higham, 2002, Lemma 3.1).

LEMMA 2.1 If $|\delta_i| \leqslant u$ and $\sigma_i = \pm 1$ for $i = 1, \ldots, k$, with $ku < 1$, then

$$\prod_{i=1}^{k} (1+\delta_i)^{\sigma_i} = 1 + \theta_k, \quad |\theta_k| \leqslant \frac{ku}{1-ku} =: \gamma_k.$$

The function γ_k defined above has two useful properties:[1]

$$\gamma_m + \gamma_n + 2\gamma_m\gamma_n \leqslant \gamma_{m+n}, \quad m, n \geqslant 0,$$

and

$$c\gamma_n \leqslant \gamma_{cn}, \quad c \geqslant 1.$$

Since we assume that $ku < 1$ for all practical k, we have

$$\gamma_k = ku + ku\gamma_k = ku + \mathcal{O}(u^2),$$

which can also be derived from the Taylor series.

[1]The two properties were listed in Higham (1997) and Higham (2002, Lemma 3.3), but with '$2\gamma_m\gamma_n$' replaced by '$\gamma_m\gamma_n$'. Here we give a slightly tighter inequality.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

These properties are used frequently to derive inequalities in this paper. Lemma 2.2 is helpful for our backward error analysis of block LDL^{T} factorization.

LEMMA 2.2 Let $y = \left(s - \sum_{k=1}^{n-1} a_k b_k c_k\right)/d$. No matter what the order of evaluation, the computed $\hat{y}$ satisfies

$$\hat{y}d + \sum_{k=1}^{n-1} a_k b_k c_k = s + \Delta s, \quad |\Delta s| \leqslant \gamma_n \left(|\hat{y}d| + \sum_{k=1}^{n-1} |a_k b_k c_k| \right).$$

Proof. The proof is similar to the derivation leading to Lemma 8.4 in Higham (2002). Let $r = s - \sum_{k=1}^{n-1} a_k b_k c_k$ and assume that it is computed in the order

$$(((s - a_1 b_1 c_1) - a_2 b_2 c_2) - \cdots - a_{n-1} b_{n-1} c_{n-1}).$$

Following the model (2.1), the computed $\hat{r}$ satisfies

$$\hat{r} = s(1+\delta_1)\cdots(1+\delta_{n-1}) - \sum_{k=1}^{n-1} a_k b_k c_k (1+\xi_k)(1+\eta_k)(1+\delta_k)\cdots(1+\delta_{n-1}),$$

where $|\delta_k|, |\xi_k|, |\eta_k| \leqslant u$. Note that each $a_k b_k c_k$ can be computed as $(a_k b_k)c_k$ or $a_k(b_k c_k)$, resulting in different values of δ_k, ξ_k and η_k, whose magnitudes, however, are all bounded by u in either case. Then

$$\hat{y} = fl(\hat{r}/d) = (\hat{r}/d)(1+\delta_n), \quad |\delta_n| \leqslant u.$$

Therefore we have $\hat{y}d = \hat{r}(1+\delta_n)$. Dividing both sides by $(1+\delta_1)\cdots(1+\delta_n)$, we obtain

$$\hat{y}d \frac{1}{(1+\delta_1)\cdots(1+\delta_n)} = s - \sum_{k=1}^{n-1} a_k b_k c_k \frac{(1+\xi_k)(1+\eta_k)}{(1+\delta_1)\cdots(1+\delta_{k-1})}.$$

By Lemma 2.1, we have

$$\hat{y}d(1+\theta_n^{(0)}) = s - \sum_{k=1}^{n-1} a_k b_k c_k (1+\theta_n^{(k)}),$$

where $|\theta_n^{(k)}| \leqslant \gamma_n$ for $k = 0, 1, \ldots, n-1$. For any order of summation, the last equation remains valid by arranging the factors $(1+\delta_i)$ for $i = 1, \ldots, n$. The result follows immediately. □

We denote the symmetric matrix under consideration by $\widetilde{A}$, which is stored as A consisting of floating-point numbers. Note that $\widetilde{A}$ can be singular, but it is unrealistic to assume that A is rank deficient because of potential rounding errors. However, in this paper we say that A is rank r if $\widetilde{A}$ is rank r.

Assuming that A is obtained by rounding the elements in $\widetilde{A}$ to the closest floating-point numbers, we have $|a_{ij} - \tilde{a}_{ij}| \leqslant u\tilde{a}_{ij}$. Therefore we have

$$|A - \widetilde{A}| \leqslant u|\widetilde{A}| \leqslant \frac{u}{1-u}|A| = \gamma_1|A| = \mathcal{O}(u).$$

The overall backward error of block LDL^{T} factorization is

$$|\widetilde{A} - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}| \leqslant |\widetilde{A} - A| + |A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}| \leqslant \gamma_1|A| + |A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}|,$$

where $\hat{L}\hat{D}\hat{L}^{\mathrm{T}}$ is the computed factorization. Since $\gamma_1|A|$ is relatively small and negligible, we consider only $|A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}|$ for simplicity.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

3. Stability analysis leading to a condition for inertia preservation

This section analyses the block LDL^{T} factorization of a full-rank symmetric matrix. The key result is that the block LDL^{T} factorization preserves the inertia numerically (i.e., in inexact arithmetic) if the matrix is not too ill-conditioned.

3.1 *Componentwise analysis*

We denote the block LDL^{T} factorization of a full-rank symmetric matrix $A \in \mathbb{R}^{n\times n}$ by LDL^{T}, where

$$D = \begin{bmatrix} B_1 & & & \\ & B_2 & & \\ & & \ddots & \\ & & & B_m \end{bmatrix}, \quad L = \begin{bmatrix} L_{11} & & & \\ L_{21} & L_{22} & & \\ \vdots & \vdots & \ddots & \\ L_{m1} & L_{m2} & \cdots & L_{mm} \end{bmatrix} \in \mathbb{R}^{n\times n}.$$

Each B_i is a 1×1 or 2×2 block, with $L_{ii} = 1$ or $L_{ii} = I_2$, respectively. The rest of L is partitioned accordingly. Algorithm 3.1 gives the block LDL^{T} factorization in inner product form.

ALGORITHM 3.1 Block LDL^{T} factorization of $A \in \mathbb{R}^{n\times n}$ in inner product form

for all $i = 1, \ldots, m$ **do** ▷ m is the number of diagonal blocks
  for all $j = 1, \ldots, i-1$ **do**
    $L_{ij} := (A_{ij} - \sum_{k=1}^{j-1} L_{ik} B_k L_{jk}^{\mathrm{T}}) B_j^{-1}$ ▷ (*)
  end for
  $B_i := A_{ii} - \sum_{k=1}^{i-1} L_{ik} B_k L_{ik}^{\mathrm{T}}$ ▷ (**)
end for

Note that, if we store all $L_{ik}B_k$ (independent of j) in an array for the computation (*) in Algorithm 3.1, then the factorization requires only $n^3/3$ flops, the same as the Cholesky factorization. The pivoting cost depends on the pivoting strategy, taking at least $\mathcal{O}(n^2)$ and up to $n^3/6$ comparisons. See, for example, Appendix A for details. Due to pivoting, block LDL^{T} factorization in practice is implemented in outer product form (1.1), such as ssytf2.f (with Bunch–Kaufman pivoting) in LAPACK (Anderson *et al.*, 1999). We begin with our stability analysis of the block LDL^{T} factorization in inner product form presented in Algorithm 3.1 and will show that the result remains valid for the outer product formulation.

In Algorithm 3.1, each multiplication by B_j^{-1} in $(*)$ with $B_j \in \mathbb{R}^{2\times 2}$ can be computed by solving a 2×2 linear system, denoted by $Ey = z$ (i.e., E is B_j). We assume that the linear system is solved successfully with computed $\hat{y}$ satisfying

$$|\Delta E| \leqslant \hat{\gamma}_c |E|, \quad \hat{\gamma}_c = \gamma_c + \mathcal{O}(u^2), \quad (E + \Delta E)\hat{y} = z \tag{3.1}$$

for some constant c, where γ_c is defined in Lemma 2.1. Note that, strictly, we include the second-order term in the bound. For convenience, it is written as $\hat{\gamma}_c = \gamma_c + \mathcal{O}(u^2)$.

The pivoting algorithms control the conditioning of each $E \in \mathbb{R}^{2\times 2}$, so the bound (3.1) with a constant c can be obtained. Higham (1997, Section 4.1) proved that, using the Bunch & Kaufman (1977) pivoting or a variant by Sorensen and Van Loan (Dongarra *et al.*, 1991, Section 5.3.2), if the system is

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

solved by Gaussian elimination with partial pivoting (GEPP), then (3.1) holds with $c = 12$, and if it is solved by explicit inverse of E with scaling

$$y = \frac{1}{e_{21}(\frac{e_{11}}{e_{21}} \cdot \frac{e_{22}}{e_{21}} - 1)} \begin{bmatrix} \frac{e_{22}}{e_{21}} & -1 \\ -1 & \frac{e_{11}}{e_{21}} \end{bmatrix} z, \quad E = \begin{bmatrix} e_{11} & e_{21} \\ e_{21} & e_{22} \end{bmatrix}, \tag{3.2}$$

as implemented in both LAPACK (Anderson *et al.*, 1999) and LINPACK (Dongarra *et al.*, 1979), then the bound is still attained with $c = 180$. The pivoting strategies of Bunch & Parlett (1971), fast Bunch–Parlett and bounded Bunch–Kaufman (Duff & Reid, 1982, 1983; Ashcraft *et al.*, 1998) also guarantee (3.1) since they all satisfy the required conditions listed in Higham (1997, Section 4.1). The bounds still hold with the same pivoting algorithms (but a different pivoting argument α; see Appendix A) in the case of symmetric triadic matrices studied by Fang & O'Leary (2006).

For block LDL^{T} factorization of a symmetric tridiagonal matrix, Higham (1999) showed that with the pivoting method of Bunch (1974, Section 4), if a 2×2 linear system is solved by GEPP, then (3.1) holds with $c = 6\sqrt{5}$. This bound, with the same derivation, is still valid for the simplified strategy of Bunch & Marcia (2006). The pivoting strategy of Bunch & Marcia (2005), for stability reasons, uses two different ways to solve 2×2 linear systems, with an analysis leading to the bound (3.1) with $c = 500$. In a similar discussion, the bound (3.1) with $c = 500$ still holds for 2×2 linear systems solved by the explicit inverse with scaling (3.2) using the simplified strategy (Bunch & Marcia, 2006), Bunch's method (Bunch, 1974, Section 4) and Higham's modification (Higham, 1999, p. 183).

We conclude that all of the pivoting strategies (Bunch & Parlett, 1971; Bunch, 1974; Bunch & Kaufman, 1977; Duff & Reid, 1982, 1983; Ashcraft *et al.*, 1998; Bunch & Marcia, 2005, 2006; Fang & O'Leary, 2006) satisfy condition (3.1), with different c summarized in Table 1.

Theorem 3.3 gives a componentwise backward error analysis of block LDL^{T} factorization, with a proof using Lemma 3.2.

LEMMA 3.2 Let $Y = \left(S - \sum_{k=1}^{m-1} A_k B_k C_k\right) E^{-1}$, where E and each B_k are either 1×1 or 2×2, such that the matrix operations are well defined. If E is a 2×2 matrix then we assume that condition (3.1) holds. Then, no matter what the order of evaluation, the computed $\hat{Y}$ satisfies

$$\hat{Y} E + \sum_{k=1}^{m-1} A_k B_k C_k = S + \Delta S, \tag{3.3}$$

TABLE 1 *The constant c in the backward error bound* (3.1) *of* 2×2 *linear systems*

		2×2 system solved by	
Matrix type	Pivoting strategy	GEPP	Inverse
General symmetric	Bunch–Kaufman Bounded Bunch–Kaufman/fast Bunch–Parlett Bunch–Parlett	$c = 12$	$c = 180$
Symmetric tridiagonal	Bunch's method Bunch–Marcia (simplified strategy)	$c = 6\sqrt{5}$	$c = 500$

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

where

$$|\Delta S| \leqslant \max\{\hat{\gamma}_c, \gamma_{4m-3}\}\left(|S| + |\hat{Y}||E| + \sum_{k=1}^{m-1} |A_k||B_k||C_k|\right). \tag{3.4}$$

If E is an identity then $\max\{\hat{\gamma}_c, \gamma_{4m-3}\}$ can be replaced by γ_{4m-3} since $\hat{\gamma}_c = 0$.

Proof. Let $Z = S - \sum_{k=1}^{m-1} A_k B_k C_k$ and hence $Y = ZE^{-1}$. If B_k is a 2×2 matrix then each element in $A_k B_k C_k$ can be represented in the form $\sum_{i=1}^{4} a_i b_i c_i$. Therefore each element in $\sum_{k=1}^{m-1} A_k B_k C_k$ sums at most $4(m-1)$ terms. By Lemma 2.2, irrespective of the order of evaluation, we have

$$\hat{Z} = Z + \Delta Z, \quad |\Delta Z| \leqslant \gamma_{4m-3}\left(|S| + \sum_{k=1}^{m-1} |A_k||B_k||C_k|\right). \tag{3.5}$$

We use $X^{(i)}$ to denote the ith row of matrix X. If E is a 2×2 matrix then applying (3.1) gives

$$\hat{Y}^{(i)}(E + \Delta E_i) = \hat{Z}^{(i)}, \quad |\Delta E_i| \leqslant \hat{\gamma}_c |E| \tag{3.6}$$

for $i = 1$ and $i = 2$ (if any). Considering ΔS in (3.3), we have

$$|\Delta S| = \left|\hat{Y}E - \left(S - \sum_{k=1}^{m-1} A_k B_k C_k\right)\right| = |\hat{Y}E - \hat{Z} + \hat{Z} - Z| = |\hat{Y}E - \hat{Z} + \Delta Z|.$$

By (3.5) and (3.6), we have

$$\begin{aligned}
|\Delta S^{(i)}| &= |-\hat{Y}^{(i)}\Delta E_i + \Delta Z^{(i)}| \leqslant |\hat{Y}^{(i)}||\Delta E_i| + |\Delta Z^{(i)}| \\
&\leqslant \hat{\gamma}_c |\hat{Y}^{(i)}||E| + \gamma_{4m-3}\left(|S^{(i)}| + \sum_{k=1}^{m-1} |A_k^{(i)}||B_k||C_k|\right) \\
&\leqslant \max\{\hat{\gamma}_c, \gamma_{4m-3}\}\left(|S^{(i)}| + |\hat{Y}^{(i)}||E| + \sum_{k=1}^{m-1} |A_k^{(i)}||B_k||C_k|\right).
\end{aligned}$$

Combining the cases $i = 1, 2$ (if any), we obtain (3.4) for E being 2×2.

If E is 1×1 then we simply apply Lemma 2.2 and obtain

$$|\Delta S| \leqslant \gamma_{4m-3}\left(|\hat{Y}||E| + \sum_{k=1}^{m-1} |A_k||B_k||C_k|\right). \qquad \square$$

THEOREM 3.3 Let $\hat{L}\hat{D}\hat{L}^{\mathrm{T}}$ be the computed block LDL^{T} factorization of a symmetric matrix $A \in \mathbb{R}^{n\times n}$. Assuming that condition (3.1) holds for all linear systems involving 2×2 pivots during the factorization, we have

$$|A - \hat{L}\hat{D}\hat{L}^{T}| \leqslant \max\{\hat{\gamma}_c, \gamma_{4m-3}\}(|A| + |\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}|), \tag{3.7}$$

where m is the number of blocks in $\hat{D}$, with $m \leqslant n$.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

Proof. We first prove (3.7) for block LDL^{T} factorization in inner product form and then show that it remains valid for the outer product form.

Applying Lemma 3.2 to $(*)$ and $(**)$ in Algorithm 3.1, which presents the inner product formulation of block LDL^{T} factorization, we obtain

$$\left| A_{ij} - \hat{L}_{ij}\hat{B}_j - \sum_{k=1}^{j-1} \hat{L}_{ik}\hat{B}_k\hat{L}_{jk}^{\mathrm{T}} \right| \leqslant \max\{\hat{\gamma}_c, \gamma_{4j-3}\} \left(|A_{ij}| + |\hat{L}_{ij}||\hat{B}_j| + \sum_{k=1}^{j-1} |\hat{L}_{ik}||\hat{B}_k||\hat{L}_{jk}^{\mathrm{T}}| \right).$$

Therefore we have

$$\left| A_{ij} - \sum_{k=1}^{j} \hat{L}_{ik}\hat{B}_k\hat{L}_{jk}^{\mathrm{T}} \right| \leqslant \max\{\hat{\gamma}_c, \gamma_{4j-3}\} \left(|A_{ij}| + \sum_{k=1}^{j} |\hat{L}_{ik}||\hat{B}_k||\hat{L}_{jk}^{\mathrm{T}}| \right) \tag{3.8}$$

for $1 \leqslant j \leqslant i \leqslant m$, where $\hat{L}_{jj} = 1$ or I_2, depending on whether B_j is 1×1 or 2×2 for $i = 1, \ldots, m$. The result (3.7) is obtained by aggregating terms in (3.8) into one matrix form, *irrespective of the order of evaluation* in (*) and (**) in Algorithm 3.1.

For the outer product form each Schur complement (1.1) can be computed by either

$$\bar{A} = A_{22} - A_{21}A_{11}^{-1}A_{21}^{\mathrm{T}} \tag{3.9}$$

or

$$\bar{A} = A_{22} - \bar{L}A_{11}\bar{L}^{\mathrm{T}}, \quad \bar{L} = A_{21}A_{11}^{-1}. \tag{3.10}$$

We now consider Algorithm 3.1, the block LDL^{T} factorization in inner product form. If we compute (*) in the order

$$L_{ij} = (((A_{ij} - L_{i1}B_1L_{j1}^{\mathrm{T}}) - L_{i2}B_2L_{j2}^{\mathrm{T}}) - \cdots - L_{i,j-1}B_{j-1}L_{j,j-1})B_j^{-1} \tag{3.11}$$

and (**) in the order

$$B_i = ((A_{ii} - L_{i1}B_1L_{i1}^{\mathrm{T}}) - L_{i2}B_2L_{i2}^{\mathrm{T}}) - \cdots - L_{i,i-1}B_{i-1}L_{i,i-1}, \tag{3.12}$$

then this inner product formulation is computationally equivalent (including equivalence of rounding errors) to the outer product formulation computed by (3.10). Note that we have utilized the following property. In each Schur complement computation by (3.10), $\bar{L}A_{11}\bar{L}^{\mathrm{T}}$ can be computed in the order $(\bar{L}A_{11})\bar{L}^{\mathrm{T}}$ or $\bar{L}(A_{11}\bar{L}^{\mathrm{T}})$. In either way we can match the computation by computing the corresponding $L_{ik}B_kL_{jk}^{\mathrm{T}}$ in (*) as $(L_{ik}B_k)L_{jk}^{\mathrm{T}}$ or $L_{ik}(B_kL_{jk}^{\mathrm{T}})$ and the corresponding $L_{ik}B_kL_{ik}^{\mathrm{T}}$ in (**) as $(L_{ik}B_k)L_{ik}^{\mathrm{T}}$ or $L_{ik}(B_kL_{ik}^{\mathrm{T}})$. As a result, the rounding errors in both formulations are identical, and therefore (3.7) remains valid for block LDL^{T} factorization in outer product form computed by (3.10).

The formula (3.9) computationally involves fewer arithmetic operations than (3.10) and therefore will not affect the validity of (3.7). To be precise, we consider (3.9) and (3.10) computed in the order

$$A_{22} - (A_{21}A_{11}^{-1})\underline{A_{21}^{\mathrm{T}}}, \quad A_{22} - (A_{21}A_{11}^{-1})(\underline{A_{11}(A_{21}A_{11}^{-1})^{\mathrm{T}}}),$$

respectively. The underlined term $A_{11}(A_{21}A_{11}^{-1})^{\mathrm{T}}$, when it is computed in inexact arithmetic, is indeed A_{21}^{T} with perturbations due to rounding errors. The discussion will be similar if (3.9) and (3.10) are

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

computed in the other order. Now let us treat rounding errors as perturbations. The extra perturbations from the computational differences between (3.9) and (3.10) accumulate iteratively while forming the block LDL^{T} factorization. Therefore the computed $\hat{L}$ and $\hat{D}$ by (3.10) allow more perturbations than those by (3.9). As a result, a backward error bound of the block LDL^{T} factorization in outer product form by (3.10) is still valid for that by (3.9). We conclude that (3.7) is guaranteed for block LDL^{T} factorization in outer product form with each Schur complement computed by either (3.9) or (3.10). □

Theorem 3.3 coincides with a result of Higham (1997, equation (4.9)), but with $\max\{\hat{\gamma}_c, \gamma_{4m-3}\}$ replaced by $p(n)u + \mathcal{O}(u^2)$, where $p(n)$ is a linear polynomial. An analogous result for block LDL^{T} factorization with Bunch–Parlett pivoting also appeared in Slapničar (1998, Theorem 7.1), but with a coefficient worse than $\max\{\hat{\gamma}_c, \gamma_{4m-3}\}$. Theorem 3.3 gives a tighter bound than the previous work.

The analysis of Higham (1997, equation (4.9)) leads to a componentwise backward error bound for solving symmetric linear systems by block LDL^{T} factorization (Higham, 1997, Theorem 4.1). Similarly, from Theorem 3.3 we can get a corresponding result. See Fang (2006, Theorem 3.4) for details.

COROLLARY 3.4 Let $\hat{L}\hat{D}\hat{L}^{\mathrm{T}}$ be the computed block LDL^{T} factorization of a symmetric tridiagonal matrix $A \in \mathbb{R}^{n\times n}$ with Bunch's method or the Bunch–Marcia pivoting. Then

$$|A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}| \leqslant \hat{\gamma}_c(|A| + |\hat{L}||\hat{D}||\hat{L}\mathrm{T}|),$$

where $\hat{\gamma}_c$ is from condition (3.1), with the values of c displayed in Table 1.

Proof. The proof is similar to that of Theorem 3.3. The differences are as follows. With both Bunch's method and Bunch–Marcia pivoting, no interchanges of rows/columns are done. Therefore the unit lower triangular matrix is indeed block bidiagonal. Then, in Algorithm 3.1, $L_{ij} = 0$ for $j < i - 1$. So we can replace the whole for loop (*) by $L_{i,i-1} = A_{i,i-1}B_{i-1}^{-1}$ and (**) by $B_i = A_{ii} - L_{i,i-1}B_iL_{i,i-1}^{\mathrm{T}}$. Now there are at most two terms in each of (*) and (**). In a discussion much the same as that leading to Theorem 3.3, we can substitute $m = 2$ into (3.7) and obtain the coefficient $\max\{\hat{\gamma}_c, \gamma_5\} = \hat{\gamma}_c$. The result follows immediately. The last equality is because, as shown in Table 1, $c = 6\sqrt{5}$ when the 2×2 linear systems are solved by GEPP and $c = 500$ when the 2×2 linear systems are solved by the explicit inverse with scaling (3.2). Both values are larger than 5. □

Corollary 3.1 coincides with a result of Higham (1999, Section 3). From Corollary 3.1 we can also obtain a componentwise backward error bound for solving symmetric linear systems by block LDL^{T} factorization. See Higham (1999, Theorem 3.2) for details.

3.2 *Normwise analysis*

Both Theorem 3.3 and Corollary 3.4 expect a bound on $\||\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}|\|$ relative to $\|A\|$. A standard forward error analysis leads to

$$|\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}| = |L||D||L^{\mathrm{T}}| + \mathcal{O}(u). \tag{3.13}$$

Therefore we can consider $\||L||D||L^{\mathrm{T}}|\|$ instead of $\||\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}|\|$ for simplicity. See also Higham (1997, Section 4.3) and Bunch & Marcia (2005, Section 3.4) for utilization of this property. Now the objective is to find a modest $c(n)$ such that

$$\||L||D||L^{\mathrm{T}}|\|_{\mathrm{M}} \leqslant c(n)\|A\|_{\mathrm{M}}, \tag{3.14}$$

where $\|\cdot\|_{\mathrm{M}}$ is the largest magnitude element in the given matrix. One may use other matrix norms, but here the maximum magnitude element norm is convenient.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

Higham (1997) showed that, using Bunch–Kaufman pivoting with the pivoting argument $\alpha = \frac{1+\sqrt{17}}{8} \approx 0.640$, the block LDL^{T} factorization of a symmetric matrix $A \in \mathbb{R}^{n\times n}$ satisfies

$$\begin{aligned} \||L||D||L^{\mathrm{T}}|\|_{\mathrm{M}} &\leqslant \max\left\{\frac{1}{\alpha}, \frac{(3+\alpha^2)(3+\alpha)}{(1-\alpha^2)^2}\right\} n\rho_n \|A\|_{\mathrm{M}} \\ &< 35.675 n\rho_n \|A\|_{\mathrm{M}}, \end{aligned} \tag{3.15}$$

where ρ_n is the growth factor, the ratio of the largest magnitude element in Schur complements to the largest magnitude element in the given matrix. More precisely, we have

$$\rho_n = \frac{\max_{i,j,k} |a_{ij}^{(k)}|}{\max_{i,j} |a_{ij}|}, \tag{3.16}$$

where the elements $a_{ij}^{(k)}$ are from the Schur complement after processing k rows/columns of the given matrix A. Note that k can be zero, in which case $a_{ij}^{(0)} = a_{ij}$.

The Bunch–Parlett, fast Bunch–Parlett and bounded Bunch–Kaufman pivoting strategies also guarantee (3.15) since they satisfy the required conditions listed in Higham (1997, Section 4.1). The inequality (3.15) still holds for the suggested pivoting argument $\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618$ for symmetric triadic matrices (Fang & O'Leary, 2006). Here we intentionally keep the coefficient 35.675 not rounded to 36 for the proof of Theorem 3.6.

For block LDL^{T} factorization of a symmetric tridiagonal matrix, Higham (1999) proved, using Bunch's method with pivoting argument $\alpha = \frac{\sqrt{5}-1}{2} \approx 0.618$, that

$$\||L||D||L^{\mathrm{T}}|\|_{\mathrm{M}} \leqslant (7\alpha + 11)(\alpha + 2)\|A\|_{\mathrm{M}} < 40.125\|A\|_{\mathrm{M}}, \tag{3.17}$$

where A is symmetric tridiagonal. Bunch & Marcia (2005, 2006) showed that their pivoting method also ensures the same bound (3.17). For the proof of Corollary 3.7 we intentionally write the coefficient 40.125 rather than 42 as in the literature (Higham, 1999; Bunch & Marcia, 2005, 2006).

We conclude that all of the pivoting strategies for block LDL^{T} factorization (Bunch & Parlett, 1971; Bunch, 1974; Bunch & Kaufman, 1977; Duff & Reid, 1982, 1983; Ashcraft *et al.*, 1998; Bunch & Marcia, 2005, 2006; Fang & O'Leary, 2006) satisfy condition (3.14).

Wilkinson (1968) showed that the Cholesky factorization of a symmetric positive definite matrix $A \in \mathbb{R}^{n\times n}$ is guaranteed to run to completion if

$$20n^{3/2}\kappa_2(A)u \leqslant 1,$$

where $\kappa_2(A) = \|A\|_2\|A^{-1}\|_2$ is the condition number of A.

Theorem 3.6 gives a sufficient condition for the success of block LDL^{T} factorization preserving the inertia (i.e., the computed block diagonal matrix $\hat{D}$ has the same inertia as the given matrix A). The proof utilizes a theorem of Weyl (Stewart & Sun, 1990, Corollary 4.9). The corresponding condition for symmetric tridiagonal matrices is given in Corollary 3.7.

THEOREM 3.5 (Weyl) Let A and B be two $n\times n$ Hermitian matrices and $\lambda_k(A)$, $\lambda_k(B)$ and $\lambda_k(A+B)$ be the eigenvalues of A, B and $A+B$ arranged in increasing order for $k = 1, \ldots, n$. Then, for $k = 1, \ldots, n$, we have

$$\lambda_k(A) + \lambda_1(B) \leqslant \lambda_k(A+B) \leqslant \lambda_k(A) + \lambda_n(B).$$

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

THEOREM 3.6 With the Bunch–Parlett, Bunch–Kaufman, fast Bunch–Parlett or bounded Bunch–Kaufman pivoting strategy, the block LDL^{T} factorization of a symmetric matrix $A \in \mathbb{R}^{n\times n}$ preserves the inertia if $f(n)\kappa_2(A) < 1$ (i.e., A is not too ill-conditioned), where

$$f(n) = 36n^2 \max\{c, 4m-3\}\rho_n u + \mathcal{O}(u^2),$$

with m being the number of diagonal blocks ($m \leqslant n$), the constant c from (3.1) and ρ_n being the growth factor defined in (3.16). The values of c are displayed in Table 1.

Proof. The proof is by induction. Let

$$A + \Delta A = \hat{L}\hat{D}\hat{L}^{\mathrm{T}}, \quad \hat{A}_k = A_k + \Delta A_k,$$

where $A_k = A(1:k, 1:k)$ and $\Delta A_k = \Delta A(1:k, 1:k)$. We now consider Algorithm 3.1, the inner product form of block LDL^{T} factorization. The process is to iteratively factor A_k, increasing k from 1 to n. The outer product form is also an iterative process by (1.1). The success of processing the first $k-s$ columns/rows implies the success of factoring A_k, where $s = 1$ or 2 denotes whether the last pivot is 1×1 or 2×2. Therefore we can concentrate on the factorization of A_k, increasing k from 1 to n.

Obviously, the first stage of the block LDL^{T} factorization succeeds numerically for a nonsingular matrix. Now suppose that the factorization of A_{k-s} is successfully completed and preserves the inertia (i.e., the inertia of $\hat{A}_{k-s}$ is the same as that of A_{k-s}), where $s = 1$ or 2 denotes whether the next pivot is 1×1 or 2×2. Since $\hat{A}_{k-s}$ preserves the inertia of A_{k-s}, all of the pivots in factoring $\hat{A}_{k-s}$ are full rank, and so the factorization of A_k succeeds (i.e., with no division by zero). The rest of this proof is to show that the inertia is preserved in $\hat{A}_k$ so the induction can be continued.

We will use the following useful property of norms (Higham, 2002, Section 6). For any matrix $H \in \mathbb{R}^{k\times k}$ we have

$$\|H\|_{\mathrm{M}} \leqslant \|H\|_2 \leqslant \|H\|_F \leqslant k\|H\|_{\mathrm{M}}. \tag{3.18}$$

Let $L_k = L(1:k, 1:k)$, $D_k = D(1:k, 1:k)$, $\hat{L}_k = \hat{L}(1:k, 1:k)$ and $\hat{D}_k = \hat{D}(1:k, 1:k)$. By Theorem 3.3 and (3.13), the componentwise backward error satisfies

$$\begin{aligned}|\Delta A_k| &\leqslant \max\{\hat{\gamma}_c, \gamma_{4m(k)-3}\}(|A_k| + |\hat{L}_k||\hat{D}_k||\hat{L}_k^{\mathrm{T}}|)\\ &= \max\{c, 4m(k)-3\}u(|A_k| + |L_k||D_k||L_k^{\mathrm{T}}|) + \mathcal{O}(u^2)\end{aligned}$$

for all possible $1 < k \leqslant n$, where $m(k) \leqslant k$ is the number of diagonal blocks in D_k. By (3.15) and (3.18), we have

$$\begin{aligned}\|\Delta A_k\|_2 &< (1 + 35.675k^2)\max\{c, 4m(k)-3\}\rho_k u\|A_k\|_2 + \mathcal{O}(u^2)\\ &< 36k^2 \max\{c, 4m(k)-3\}\rho_k u\|A_k\|_2 + \mathcal{O}(u^2) =: f(k)\|A_k\|_2\end{aligned}$$

for $k > 1$. The case $k = 1$ (if any) is trivially valid.

Now let

$$\lambda_*(A_k) = \min_{1\leqslant i\leqslant k} |\lambda_i(A_k)|.$$

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

TABLE 2 *Bounds on growth factor ρ_n defined in* (3.16)

Symmetric matrix	General	Triadic
Pivoting argument	$\alpha = \frac{1+\sqrt{17}}{8}$	$\alpha = \frac{\sqrt{5}-1}{2}$
Bunch–Kaufman	2.57^{n-1}	$\mathcal{O}(1.28^n)$
Bounded Bunch–Kaufman/fast Bunch–Parlett		$\mathcal{O}(n^{1.7})$
Bunch–Parlett	$5.4n^{1+(\ln n)/4}$	

By Theorem 3.5, if $\lambda_i(A_k) > 0$ then we have

$$\begin{aligned}\lambda_i(A_k + \Delta A_k) &\geqslant \lambda_i(A_k) - \|\Delta A_k\|_2 \\ &> \lambda_*(A_k) - f(k)\|A_k\|_2 \\ &= \lambda_*(A_k)(1 - f(k)\kappa_2(A_k)) \\ &\geqslant \lambda_*(A)(1 - f(n)\kappa_2(A)) > 0,\end{aligned}$$

where we have used the condition $f(n)\kappa_2(A) < 1$ for the last inequality. Under the same condition, if $\lambda_i(A_k) < 0$ then we have

$$\lambda_i(A_k + \Delta A_k) \leqslant \lambda_i(A_k) + \|\Delta A_k\|_2 < -\lambda_*(A_k) + f(k)\|A_k\|_2 < 0.$$

We conclude that, if $f(n)\kappa_2(A) < 1$, then $\lambda_i(A_k + \Delta A_k)$ and $\lambda_i(A_k)$ have the same sign. In other words, $\hat{A}_k$ preserves the inertia of A_k. By induction, the condition $f(n)\kappa_2(A) < 1$ guarantees that the computed block LDL^{T} factorization preserves the inertia of A. □

Table 2 displays the bounds on the growth factor ρ_n, collected from the literature. The bounds for general symmetric matrices are from Bunch (1971) and Bunch & Kaufman (1977). The bounds for symmetric triadic matrices are from Fang & O'Leary (2006).

Sorensen and Van Loan (Dongarra *et al.*, 1991, Section 5.3.2) suggested a variant of the Bunch–Kaufman pivoting strategy. See also Appendix A.2. Higham (1997) indicated that this variant preserves the properties (3.1) and (3.15). Therefore Theorem 3.6, as well as the bounds in Table 2, remains valid.

COROLLARY 3.7 Using Bunch's method or the Bunch–Marcia pivoting, the block LDL^{T} factorization of a symmetric tridiagonal matrix $A \in \mathbb{R}^{n\times n}$ preserves the inertia of A if $f(n)\kappa_2(A) < 1$, where

$$f(n) = 42cnu + \mathcal{O}(u^2),$$

with the constant c from (3.1). The values of c are displayed in Table 1.

Proof. The proof is much the same as that of Theorem 3.6, but using Corollary 3.4 rather than Theorem 3.3, and invoking (3.17) instead of (3.15). □

The meaning of Theorem 3.6 and Corollary 3.7 is that the block LDL^{T} factorization preserves the inertia if the matrix is not too ill-conditioned.

4. Stability analysis for the reliability of rank estimation and inertia estimation

The block LDL^{T} factorization, at least in exact arithmetic, with the Bunch–Parlett, fast Bunch–Parlett or bounded Bunch–Kaufman pivoting strategy is inertia revealing. Therefore it can be applied to inertia

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

estimation for symmetric matrices. This section analyses the stability of the block LDL^{T} factorization of a rank-deficient matrix. The result, supported by the numerical evidence in Section 5, indicates that its application to rank estimation and inertia estimation is reliable.

The direction of this section was inspired by the work of Higham (1990) on the stability analysis of the Cholesky factorization with complete pivoting for symmetric semidefinite matrices.

4.1 *Componentwise analysis*

We consider the block LDL^{T} factorization of a rank-deficient symmetric matrix $A \in \mathbb{R}^{n\times n}$ of rank $r < n$. Without loss of generality, we assume that the necessary interchanges are done so that $A(1:r, 1:r)$ has full rank r, as they would be with the Bunch–Parlett, fast Bunch–Parlett or bounded Bunch–Kaufman pivoting. The factorization is denoted by $A = LDL^{\mathrm{T}}$, where

$$D = \begin{bmatrix} B_1 & & & \\ & B_2 & & \\ & & \ddots & \\ & & & B_{m-1} \end{bmatrix} \in \mathbb{R}^{r\times r}, \quad L = \begin{bmatrix} L_{11} & & & \\ L_{21} & L_{22} & & \\ \vdots & \vdots & \ddots & \\ L_{m-1,1} & L_{m-1,2} & \cdots & L_{m-1,m-1} \\ L_{m1} & L_{m2} & \cdots & L_{m,m-1} \end{bmatrix} \in \mathbb{R}^{n\times r}.$$

Each B_i is either 1×1 or 2×2, with $L_{ii} = 1$ or $L_{ii} = I_2$, respectively. A pseudocode is given in Algorithm 4.1.

ALGORITHM 4.1 Block LDL^{T} factorization of $A \in \mathbb{R}^{n\times n}$ of rank $r < n$

for all $j = 1, \ldots, m$ **do** ▷ The number of diagonal blocks in $A(1:r, 1:r)$ is $m-1$
  $B_j := A_{jj} - \sum_{k=1}^{j-1} L_{jk} B_k L_{jk}^{\mathrm{T}}$ ▷ (**)
  for all $i = j+1, \ldots, m$ **do**
    $L_{ij} := (A_{ij} - \sum_{k=1}^{j-1} L_{ik} B_k L_{jk}^{\mathrm{T}}) B_j^{-1}$ ▷ (*)
  end for
end for

Algorithm 4.1 is indeed the incomplete block LDL^{T} factorization of A after processing r rows/columns, no matter what the rank of A is. Theorem 4.2 gives a bound on its componentwise backward error. In inertia estimation we are concerned with a symmetric matrix $A \in \mathbb{R}^{n\times n}$ of rank $r < n$.

THEOREM 4.2 Consider the incomplete block LDL^{T} factorization of a symmetric matrix $A \in \mathbb{R}^{n\times n}$ after processing r rows/columns ($r < n$). Assume that condition (3.1) holds for all linear systems involving 2×2 pivots, and define the backward error ΔA by

$$A + \Delta A = \hat{L}\hat{D}\hat{L}^{\mathrm{T}} + \hat{A}^{(r+1)}, \tag{4.1}$$

where

$$\hat{A}^{(r+1)} = \begin{array}{c} \\ r \\ n-r \end{array} \begin{array}{c} \begin{array}{cc} r & n-r \end{array} \\ \begin{bmatrix} 0 & 0 \\ 0 & \hat{S}_r(A) \end{bmatrix} \end{array},$$

with $\hat{S}_r(A)$ being the computed Schur complement. Then

$$|\Delta A| \leqslant \max\{\hat{\gamma}_c, \gamma_{4r+1}\}(|A| + |\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}| + |\hat{A}^{(r+1)}|). \tag{4.2}$$

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

Proof. We first prove (4.2) for the incomplete block LDL^{T} factorization computed by Algorithm 4.1, followed by a discussion to show that (4.2) remains valid for the incomplete block LDL^{T} factorization in outer product form.

Consider Algorithm 4.1. To simplify the notation, we let $\hat{L}_{ii} = 1$ or I_2 depending on whether B_i is 1×1 or 2×2 for $i = 1, \ldots, m-1$. By Lemma 3.2 with the condition (3.1), we have

$$\left| A_{ij} - \sum_{k=1}^{j} \hat{L}_{ik}\hat{B}_k\hat{L}_{jk}^{\mathrm{T}} \right| \leqslant \max\{\hat{\gamma}_c, \gamma_{4j-3}\} \left(|A_{ij}| + \sum_{k=1}^{j} |\hat{L}_{ik}||\hat{B}_k||\hat{L}_{jk}^{\mathrm{T}}| \right) \tag{4.3}$$

for $1 \leqslant j \leqslant i \leqslant m$, irrespective of the order of evaluation in (*) and (**). It leaves only the last Schur complement computed on A_{mm} not yet taken into account. Note that, although A_{mm} can be larger than 2×2, the error analysis in Lemma 3.2 for $\hat{S}_r(A)$ is still valid. Therefore we have

$$\left| A_{mm} - \hat{S}_r(A) - \sum_{k=1}^{m-1} \hat{L}_{mk}\hat{B}_k\hat{L}_{mk}^{\mathrm{T}} \right| \leqslant \max\{\hat{\gamma}_c, \gamma_{4m-3}\} \left(|A_{mm}| + |\hat{S}_r(A)| + \sum_{k=1}^{m-1} |\hat{L}_{mk}||\hat{B}_k||\hat{L}_{mk}^{\mathrm{T}}| \right). \tag{4.4}$$

The result (4.2) is obtained by collecting terms in (4.3) together with (4.4) into one matrix form.

Due to pivoting, the incomplete block LDL^{T} factorization, in practice, is computed by the outer product formulation (1.1) rather than by Algorithm 4.1. Because of computational equivalence, Algorithm 4.1 produces $\hat{L}(1:r, 1:r)$ and $\hat{D}(1:r, 1:r)$ that are the same (including rounding errors) as those from Algorithm 3.1 applied to factor $A(1:r, 1:r)$, assuming that the order of each multiplication of three terms in (*) and (**) matches the corresponding one in the other algorithm. This observation makes it possible to reuse the bridge between the inner and the outer product formulations of the block LDL^{T} factorization in floating-point computation in the proof of Theorem 3.3.

In short, if all (*) and (**) in Algorithm 4.1 are computed in the order (3.11) and (3.12), then the result is equivalent (including the equivalence of rounding errors) to the outer product formulation with each Schur complement computed by (3.10). So (4.2) remains valid. The formula (3.9) allows less perturbations by rounding errors than (3.10) and therefore will not affect the validity of (4.2). See the proof of Theorem 3.3 for a detailed discussion. We conclude that the block LDL^{T} factorization in outer product form with each Schur complement computed by either (3.9) or (3.10) guarantees (4.2). □

Theorem 4.2 means that, if a rank-revealing block LDL^{T} factorization algorithm results in ΔA and $\hat{A}^{(r+1)}$ that are negligible compared with the given $A \in \mathbb{R}^{n\times n}$ of rank $r < n$, then it can be utilized for rank estimation.

4.2 *Normwise analysis*

Now we bound $\|A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}\|_F$ in (4.1) for analysing the stability of a rank-revealing block LDL^{T} factorization algorithm. The result will support the reliability of its application to rank estimation and inertia estimation for symmetric indefinite matrices. We begin with a perturbation theorem for block LDL^{T} factorization.

THEOREM 4.3 Let $S_k(A)$ be the Schur complement appearing in a block LDL^{T} factorization of a symmetric matrix $A \in \mathbb{R}^{n\times n}$ after processing the first k rows/columns, $k < n$. Suppose that there is a symmetric perturbation in A, denoted by F. Partition A as

$$A = \begin{bmatrix} A_{11} & A_{21}^{\mathrm{T}} \\ A_{21} & A_{22} \end{bmatrix},$$

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

where $A_{11} \in \mathbb{R}^{k\times k}$, and partition F accordingly. Assume that both A_{11} and $A_{11} + F_{11}$ are nonsingular. Then

$$\|S_k(A+F) - S_k(A)\| \leqslant (\|W\|+1)^2\|F\| + \mathcal{O}(\|F\|^2),$$

where $W = A_{11}^{-1}A_{21}^{\mathrm{T}}$, and $\|\cdot\|$ is a p-norm, ∞-norm or Frobenius norm.

Theorem 4.3 is from Fang & O'Leary (2006, Theorem 5.1), a generalization of a lemma by Higham (1990, Lemma 9.2) for the Cholesky factorization.

THEOREM 4.4 With the Bunch–Parlett, fast Bunch–Parlett or bounded Bunch–Kaufman pivoting, the block LDL^{T} factorization of a symmetric indefinite matrix $A \in \mathbb{R}^{n\times n}$ of rank r satisfies

$$\|A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}\|_F \leqslant \max\{c, 4r+1\}(\tau(A)+1)((\|W\|_F+1)^2+1)u\|A\|_F + \mathcal{O}(u^2),$$

where c is from condition (3.1), $W = A(1{:}r, 1{:}r)^{-1}A(1{:}r, r{+}1{:}n)$ and

$$\tau(A) := \||L||D||L^{\mathrm{T}}|\|_F/\|A\|_F \leqslant 36n(r+1)\rho_{r+1}, \tag{4.5}$$

with ρ_{r+1} being the growth factor defined in (3.16). Note that we have assumed that the interchanges of rows and columns by pivoting are done so that $A(1{:}r, 1{:}r)$ has full rank.

Proof. Since (3.1) holds, so does (4.2), from which we can say that $\Delta A = \mathcal{O}(u)$ if $\||\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}|\|_F$ and $\|\hat{A}^{(r+1)}\|_F$ are well bounded in terms of $\|A\|_F$. Alternatively, we can simply consider $\||L||D||L^{\mathrm{T}}|\|_F$ and $\|A^{(r+1)}\|_F$ because a standard forward error analysis based on the floating-point arithmetic model (2.1) ensures (3.13) and that

$$\hat{A}^{(r+1)} = A^{(r+1)} + \mathcal{O}(u) = \mathcal{O}(u). \tag{4.6}$$

Note that here $\|L\|_{\mathrm{M}}$ is bounded by a constant, and $\|D\|_{\mathrm{M}}/\|A\|_{\mathrm{M}}$ is no larger than the growth factor, whose bounds are listed in Table 2. See also Fang & O'Leary (2006, Section 3) for a discussion on the bounds. Hence the $\mathcal{O}(u)$ terms in both (3.13) and (4.6) are especially negligible.

Now we bound $\tau(A)$. Applying the analysis of Higham (1997, Section 4.3) leading to (3.15), we obtain

$$\||L||D||L^{\mathrm{T}}|\|_{\mathrm{M}} \leqslant 36(r+1)\rho_{r+1}\|A\|_{\mathrm{M}},$$

which implies (4.5), where we have utilized (3.18) to transform the norm. We conclude from (3.13), (4.2), (4.5) and (4.6) that

$$\Delta A = \mathcal{O}(u), \tag{4.7}$$

where ΔA is defined in (4.1).

Note that $\hat{L}\hat{D}\hat{L}^{\mathrm{T}}$ is indeed the incomplete block LDL^{T} factorization of $A + \Delta A$ with $\hat{A}^{(r+1)}$ being the Schur complement. By Theorem 4.3, we obtain the perturbation bound

$$\|\hat{A}^{(r+1)}\|_F \leqslant (\|W\|_F+1)^2\|\Delta A\|_F + \mathcal{O}(u^2), \tag{4.8}$$

where $W = A(1{:}r, 1{:}r)^{-1}A(1{:}r, r{+}1{:}n)$. By (4.1), (4.6) and (4.7), we have $\hat{L}\hat{D}\hat{L}^{\mathrm{T}} = A + \mathcal{O}(u)$. By (4.2), (4.5) and (4.6), we have

$$\begin{aligned}\|\Delta A\|_F &\leqslant \max\{\hat{\gamma}_c, \gamma_{4r+1}\}(\||\hat{L}||\hat{D}||\hat{L}^{\mathrm{T}}|\|_F + \|A\|_F + \|\hat{A}^{(r+1)}\|_F)\\ &= \max\{c, 4r+1\}(\tau(A)+1)u\|A\|_F + \mathcal{O}(u^2).\end{aligned} \tag{4.9}$$

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

Substituting (4.9) into (4.8), we obtain

$$\|\hat{A}^{(r+1)}\|_F \leqslant \max\{c, 4r+1\}(\tau(A)+1)(\|W\|_F+1)^2 u\|A\|_F + \mathcal{O}(u^2). \tag{4.10}$$

The result is concluded from (4.1), (4.9) and (4.10). □

By Theorem 4.4, the bound on $\|A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}\|_F/\|A\|_F$ is governed by $\|W\|_F$ and $\tau(A)$. A bound on $\|W\|_F$ was given in Fang & O'Leary (2006, Lemma 5.3) that

$$\|W\|_F \leqslant \sqrt{\frac{\zeta}{\zeta+2}(n-r)((1+\zeta)^{2r}-1)}, \tag{4.11}$$

where $\zeta = \max\{\frac{1}{\alpha}, \frac{1}{1-\alpha}\}$ is the bound on the elements of $|L|$. With the suggested pivoting argument, $\alpha = \frac{1+\sqrt{17}}{8} \approx 0.640$ and $\zeta = \frac{7+\sqrt{17}}{4} \approx 2.781$.

We conclude that, with Bunch–Parlett, fast Bunch–Parlett or bounded Bunch–Kaufman pivoting, block LDL^{T} factorization is normwise backward stable for rank-deficient symmetric matrices.

Two remarks are worth noting. First, we exclude the Bunch–Kaufman pivoting in Theorem 4.4 because the block LDL^{T} factorization with Bunch–Kaufman pivoting is not, in general, rank revealing. Second, block LDL^{T} factorization is more than rank revealing: it is inertia revealing. If $A(1:r, 1:r)$ satisfies the condition in Theorem 3.6, then the inertia of this nonsingular part is certainly preserved.

4.3 *Symmetric tridiagonal matrices*

The block LDL^{T} factorization of a symmetric tridiagonal matrix with Bunch's pivoting method or the Bunch–Marcia pivoting always succeeds numerically. This is different from the general symmetric case with Bunch–Kaufman pivoting that does not guarantee the completion of block LDL^{T} factorization.

When a symmetric tridiagonal matrix $A \in \mathbb{R}^{n\times n}$ is irreducible (or, equivalently, all subdiagonal elements are nonzero), Bunch's pivoting method and the Bunch–Marcia pivoting, in either exact or inexact arithmetic computation, always find a nonsingular 1×1 pivot or 2×2 block pivot in each Schur complement, except the last pivot, which can contain a zero eigenvalue if it is 1×1. See, for example, Appendix B for details. Therefore the rank of A is either n or $n-1$, determined by the last pivot of the block LDL^{T} factorization. Numerically, if the last pivot is 1×1 with magnitude smaller than some tolerance, then we say that A has rank $n-1$. Otherwise, the rank is n. Corollary 3.7 gives a condition under which the block LDL^{T} factorization is guaranteed to preserve the inertia of a given matrix.

When a given symmetric tridiagonal matrix is reducible, one can split it into multiple irreducible ones, compute the block LDL^{T} factorization of each of them and then combine the results.

The block LDL^{T} factorization, accompanied with Sylvester's law of inertia, gives a simple proof of Lemma 4.5 described as follows.

LEMMA 4.5 An irreducible symmetric tridiagonal matrix cannot have multiple eigenvalues.

Proof. Consider, in exact arithmetic, the block LDL^{T} factorization with Bunch's pivoting method or the Bunch–Marcia pivoting of a given irreducible symmetric tridiagonal matrix A. The diagonal blocks are all nonsingular except the last one, which can have a zero eigenvalue. By Sylvester's law of inertia, A has at most one zero eigenvalue.

Now suppose that we have an irreducible symmetric tridiagonal matrix A with multiple eigenvalues denoted by λ. Then $A - \lambda I$ remains symmetric tridiagonal and irreducible, but it has multiple zero eigenvalues. By contradiction, A cannot have multiple eigenvalues. □

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

Lemma 4.5 is also valid for complex Hermitian matrices with a little thought. It gives an explanation as to why the Lanczos method for computing eigenvalues (Lanczos, 1950) often encounters difficulties in practice when the given symmetric matrix has multiple eigenvalues. This is because the Lanczos algorithm approximates the eigenvalues by Ritz values, which are eigenvalues of an irreducible symmetric tridiagonal matrix that cannot have multiple eigenvalues. A more general result is the strict interlacing property of eigenvalues of an irreducible symmetric tridiagonal matrix. See, for example, Golub & Van Loan (1996, Theorem 8.5.1).

5. Experiments on rank and inertia estimation

The experiments were on a few PCs with machine epsilon $2^{-52} \approx 2.22 \times 10^{-16}$. Note that the unit roundoff u is half of the machine epsilon. Our implementation is based on the newmat library, a library of matrix operations written in C++ by Robert Davies. Each block LDL^{T} factorization was formed by outer product formulation (1.1), with the Schur complement computed by (3.10).

For both rank estimation and inertia estimation an important practical issue is when to stop the factorization. The bound (4.10) suggests a stopping criterion of the form

$$\|\hat{A}^{(k+1)}\|_F \leqslant f(n,k)u\|A\|_F, \tag{5.1}$$

where the computed Schur complement measured in the Frobenius norm serves as the residual on the left-hand side and the tolerance is on the right-hand side. Alternatively, we can use the less expensive

$$\|\hat{B}_i\|_F \leqslant f(n,k)u\|B_1\|_F, \tag{5.2}$$

where $\hat{B}_i$ is the block pivot chosen in $\hat{A}^{(k+1)}$. With the Bunch–Parlett pivoting strategy, $\|A\|_{\mathrm{M}} \approx \|B_1\|_{\mathrm{M}}$ and $\|\hat{A}^{(k+1)}\|_{\mathrm{M}} \approx \|B_i\|_{\mathrm{M}}$. Therefore (5.1) and (5.2) are related.

The term $f(n,k)$ in (5.1) takes the place of $\max\{c, 4r+1\}(\tau(A)+1)(\|W\|_F+1)^2$ in (4.10). The factors $\tau(A)$ and $\|W\|_F$ are bounded by (4.5) and (4.11), respectively. However, both bounds are too pessimistic. To investigate the ranges of $\tau(A)$ and $\|W\|_F$ of various matrices in practice, we used the random matrices described as follows.

Each indefinite matrix was constructed as $Q\Lambda Q^{\mathrm{T}} \in \mathbb{R}^{n\times n}$, where $\Lambda = \mathrm{diag}(\lambda_i)$ is of rank r, and Q, different for each matrix, is a random orthogonal matrix generated by the method of Stewart (1980), with the normal random numbers generated by the code of John Burkardt. We used three types of matrices characterized as follows:

$$|\lambda_1| = |\lambda_2| = \cdots = |\lambda_{r-1}| = 1, \quad \lambda_r = \sigma,$$
$$|\lambda_1| = |\lambda_2| = \cdots = |\lambda_{r-1}| = \sigma, \quad \lambda_r = 1,$$
$$|\lambda_i| = \beta^i, \quad i = 1,\ldots,r-1, \quad \lambda_r = 1,$$

where $0 < \sigma \leqslant 1$ and $\beta^{r-1} = \sigma$. We assigned the sign of λ_i randomly for $i = 1,\ldots,r-1$, subject to a prescribed number of negative eigenvalues, denoted by t. The value of σ governs the ambiguity of rank and inertia. Note that the test matrices were generated in a similar way to that for symmetric semidefinite matrices used by Higham (1990, Section 9.6).

Four test sets, denoted by A1, A2, B1 and B2, were used in our experiments. Each set has three groups of matrices, of the three types listed above. Set A1 contains matrices with all combinations of $n = 10, 20, \ldots, 100$, $r = 2, 3, \ldots, n$, $t = 1, 2, \ldots, r-1$ and $\sigma = 1, 10^{-3}, \ldots, 10^{-12}$ for a total of 94,875 indefinite matrices in each group. Set B1 contains matrices with all combinations of

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

$n = 100, 200, \ldots, 1000$, $r = 0.1n, 0.2n, \ldots, n$, $t = 0.2r, 0.4r, 0.6r, 0.8r$ and $\sigma = 1, 10^{-3}, \ldots, 10^{-12}$ for a total of 2000 indefinite matrices in each group. By setting $\sigma = 10^{-13}$, which is closer to machine epsilon $2^{-52} \approx 2.22 \times 10^{-16}$, we obtained matrices with ambiguous ranks for further experiments. The resulting sets of matrices corresponding to the sets A1 and B1 are denoted by A2 and B2. The characteristics of these data sets are summarized in Table 3.

Roughly speaking, sets A1 and A2 contain small size matrices, whereas sets B1 and B2 contain large size matrices. The matrices in sets A1 and B1 have clear ranks, whereas the ranks of the matrices in sets A2 and B2 are ambiguous.

In addition to $\tau(A)$ and $\|W\|_F$, we also measured the growth factor ρ_{r+1} that appeared in the bound (4.5) on $\tau(A)$, as well as the relative backward error ξ_r and the relative Schur residual η_r, defined as

$$\xi_r = \|A - \hat{L}\hat{D}\hat{L}^{\mathrm{T}}\|_F/(u\|A\|_F), \quad \eta_r = \|\hat{A}^{(r+1)}\|_F/(u\|A\|_F).$$

Their maximum values with the Bunch–Parlett, fast Bunch–Parlett and bounded Bunch–Kaufman pivoting strategies are reported in Tables 4 and 5 using the random matrices in sets A1 and B1, respectively. Note that the values of ξ_r and η_r depend on whether the 2×2 systems are solved by GEPP or explicit inverse with scaling (3.2), and so the numbers are reported separately. The values of ρ_{r+1}, $\tau(A)$ and W, by definition, are independent of the rounding errors. However, the computed values are perturbed due to roundoff.

TABLE 3 *Test sets*

Set	No. of matrices	n	r	t	σ
A1	$94{,}875 \times 3$	$10, 20, \ldots, 100$	$2, 3, \ldots, n$	$1, 2, \ldots, r-1$	$1, 10^{-3}, \ldots, 10^{-12}$
A2	$18{,}975 \times 3$				10^{-13}
B1	2000×3	$100, 200, \ldots, 1000$	$0.1n, 0.2n, \ldots, n$	$0.2r, 0.4r, \ldots, 0.8r$	$1, 10^{-3}, \ldots, 10^{-12}$
B2	400×3				10^{-13}

TABLE 4 *Experimental maximum ρ_{r+1}, $\tau(A)$, $\|W\|_F$, ξ_r and η_r of matrices in test set A1*

				2×2 GEPP		2×2 inverse	
Pivoting algorithm	ρ_{r+1}	$\tau(A)$	$\|W\|_F$	ξ_r	η_r	ξ_r	η_r
Bunch–Parlett	21.153	39.324	33.531	92.260	92.229	93.136	92.923
Fast Bunch–Parlett	21.984	42.099	40.867	206.225	206.196	261.682	261.447
Bounded Bunch–Kaufman	33.019	55.488	124.029	41,304.9	41,305.0	41,307.3	41,307.6

TABLE 5 *Experimental maximum ρ_{r+1}, $\tau(A)$, $\|W\|_F$, ξ_r and η_r of matrices in test set B1*

				2×2 GEPP		2×2 inverse	
Pivoting algorithm	ρ_{r+1}	$\tau(A)$	$\|W\|_F$	ξ_r	η_r	ξ_r	η_r
Bunch–Parlett	31.009	548.31	189.35	721.995	721.409	662.827	662.352
Fast Bunch–Parlett	33.037	554.00	175.32	755.959	755.515	734.106	733.613
Bounded Bunch–Kaufman	59.861	822.97	359.50	5815.03	5815.07	5231.91	5231.72

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

According to Table 1, using the explicit inverse with scaling (3.2) to solve 2×2 systems arising in the block LDL^{T} factorization would result in a larger backward error bound than using GEPP. However, it is too early to say that GEPP achieves better numerical stability than (3.2) for solving 2×2 systems that occur in the block LDL^{T} factorization. We can see from Table 5 that using (3.2), the explicit inverse with scaling, may produce smaller backward errors and Schur residuals than using GEPP.

One potential problem is that continuing the factorization on $\hat{A}^{(r+1)}$ is possible, i.e., not stopping through (5.1) or (5.2) when $k = r$. It affects the accuracy of the estimated rank and inertia. However, the element growth is well controlled by pivoting. The dimensions of Schur complements are reduced, whereas the right-hand sides in (5.1) and (5.2) are increased during the decomposition. These properties safeguard the reliability of the estimated rank and inertia.

All numbers reported in Tables 4 and 5 are modest, except that the bounded Bunch–Kaufman pivoting strategy introduced relatively large backward errors and Schur residuals. The values of $\tau(A)$ and $\|W\|_F$ increased moderately when the matrix size was increased. The assessment reveals that we should use a modest $f(n,k)$ in (5.1) and (5.2), perhaps subquadratic, but at least linear because of the factor $\max\{c, 4r+1\}$ in (4.10). After carrying out several experiments, we suggest $f(n,k)$ of the form

$$f(n,k) = \pi n^p (k+1)^q, \quad 1 \leqslant p+q \leqslant \frac{3}{2}, \ \ \pi > 0, \text{ a modest constant.}$$

The optimal p, q and π depend on several factors, such as ambiguity of the rank and eigenspectrum of the matrix.

Now we report the experiments using the stopping criteria (5.1) and (5.2) with $f(n,k) = \pi n$ and $f(n,k) = \pi(k+1)^{4/3}$. For the test sets A1 and A2 with matrix sizes up to $n = 100$ we used $\pi = 2, 4, \ldots, 10$. For the test sets B1 and B2 with matrix sizes up to $n = 1000$ we used $\pi = 0.5, 1, \ldots, 2.5$. Let r be the rank of a given matrix A and let $\hat{r}$ be the estimated rank. An error occurs if $\hat{r} \neq r$. The performance of rank estimation was measured by the error rate and the average error, the mean of nonzero $|\hat{r} - r|$.

Figures 1–5 show the results using various test sets A1, A2, B1 and B2. All 2×2 systems that occurred during the block LDL^{T} factorization were solved by the explicit inverse with scaling (3.2). We found that using GEPP to solve the 2×2 systems could result in slightly different estimated rank or inertia, which may or may not be an improvement. The Bunch–Parlett, fast Bunch–Parlett and bounded Bunch–Kaufman pivoting strategies are abbreviated as BP, FBP and BBK, respectively.

Accurate estimated inertia implies accurate estimated rank, but not vice versa. In practice, we found only two instances in all of the tests where the estimated rank was accurate but the estimated inertia was not. Both instances were from test set A2 using bounded Bunch–Kaufman pivoting and the stopping criterion (5.1). One used the matrix of size $n = 50$, rank $r = 42$ and number of negative values $t = 36$. The other had the matrix of size $n = 30$, rank $r = 24$ and number of negative values $t = 16$. Except for these two instances, the plots of error rates in Figs 1, 2, 4 and 5 are for both rank estimation and inertia estimation.

Figures 1 and 4 show that, for matrices in the sets A1 and B1, $f(n,k) = \pi(k+1)^{4/3}$, on average, worked better than $f(n,k) = \pi n$, especially when the economic stopping criterion (5.2) was used instead of (5.1). With appropriate parameters, the results were excellent. For example, using any of the stopping criteria (5.1) and (5.2) with $f(n,k) = \pi(k+1)^{4/3}$, we obtained all estimated ranks and inertias of matrices in set A1 accurately with $\pi = 8$ and all estimated ranks and inertias of matrices in set B1 accurately with $\pi = 1$. However, for matrices in the sets A2 and B2 with ambiguous ranks, $f(n,k) = \pi n$ was a better choice, as shown in Figs 2 and 5. Comparing Fig. 1 with Fig. 4, we can see that, on average, larger matrices resulted in higher error rates than smaller ones. As shown in Figs 2 and 5, the difference got smaller for matrices with ambiguous ranks.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

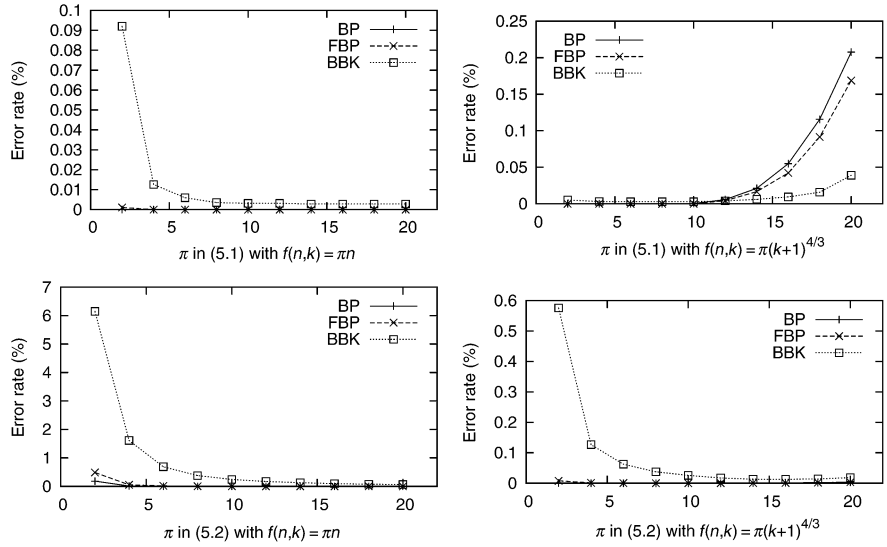

FIG. 1. Error rates using test set A1 of 284,625 indefinite matrices of size up to $n = 100$.

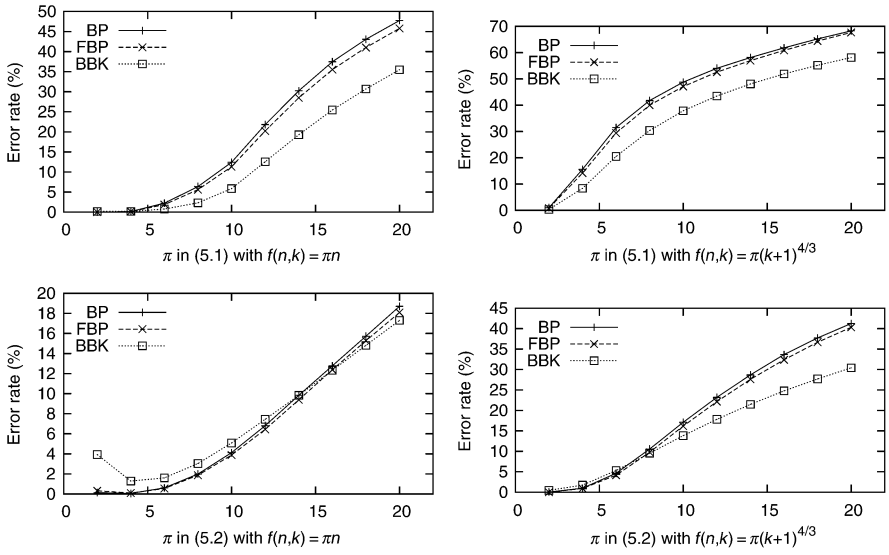

FIG. 2. Error rates using test set A2 of 56,925 indefinite matrices of sizes up to $n = 100$.

One may note that the bounded Bunch–Kaufman pivoting, with reduced cost, sometimes outperformed the Bunch–Parlett and fast Bunch–Parlett pivoting strategies in terms of error rate for matrices in the sets A2 and B2. On the other hand, such as shown in Fig. 3 for the set A2, the Bunch–Parlett and fast Bunch–Parlett pivoting strategies usually worked better in terms of average error for matrices. The plots for the test sets A1, B1 and B2 are omitted to save space.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

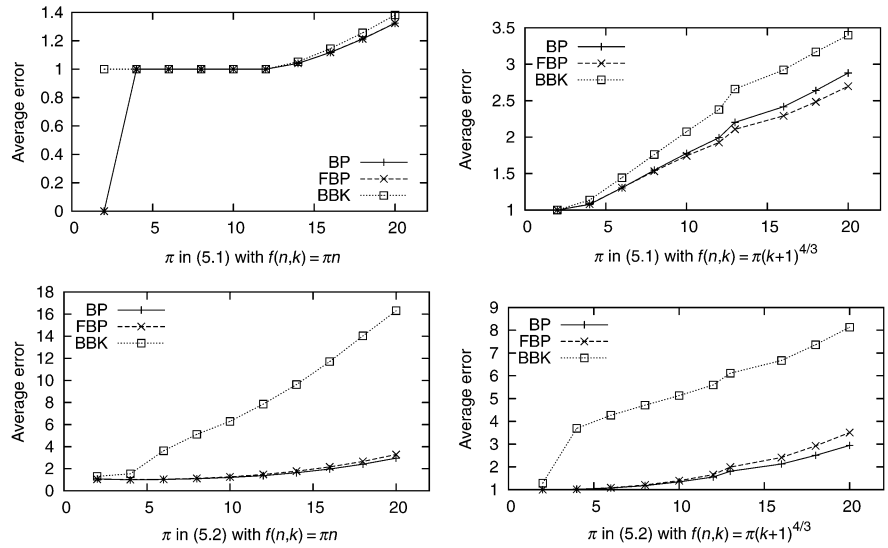

FIG. 3. Average errors using test set A2 of 56,925 indefinite matrices of sizes up to $n = 100$.

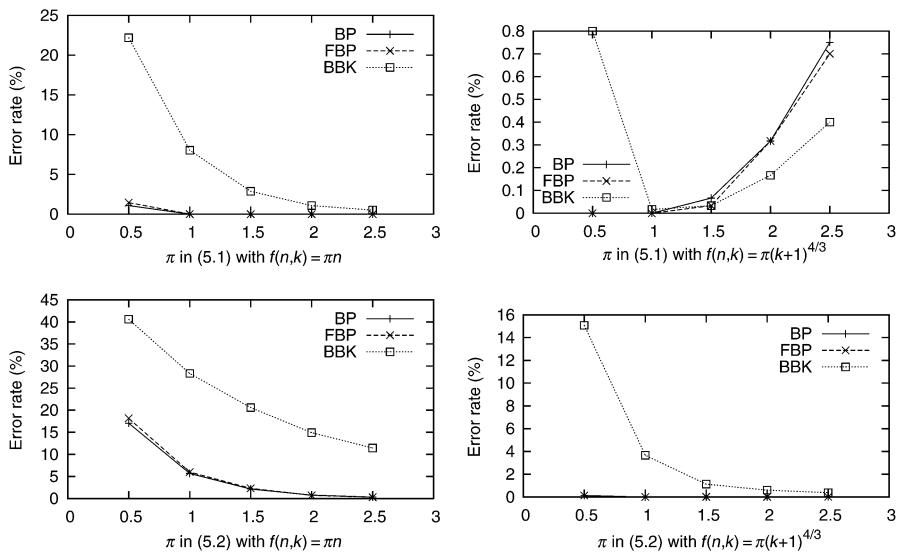

FIG. 4. Error rates using test set B1 of 6000 indefinite matrices of size up to $n = 1000$.

For symmetric positive semidefinite matrices the block LDL^{T} factorization with Bunch–Parlett or fast Bunch–Parlett pivoting reduces to the Cholesky factorization with complete pivoting in $\bar{L}\bar{D}\bar{L}^{\mathrm{T}}$ form, where $\bar{L}$ is unit lower triangular and $\bar{D}$ is diagonal (i.e., all pivots are 1×1). Indeed, for symmetric semidefinite matrices the Bunch–Parlett and fast Bunch–Parlett algorithms determine the same pivots. However, the fast Bunch–Parlett pivoting is much more efficient for such matrices since it traverses only the diagonal and one column in each Schur complement.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

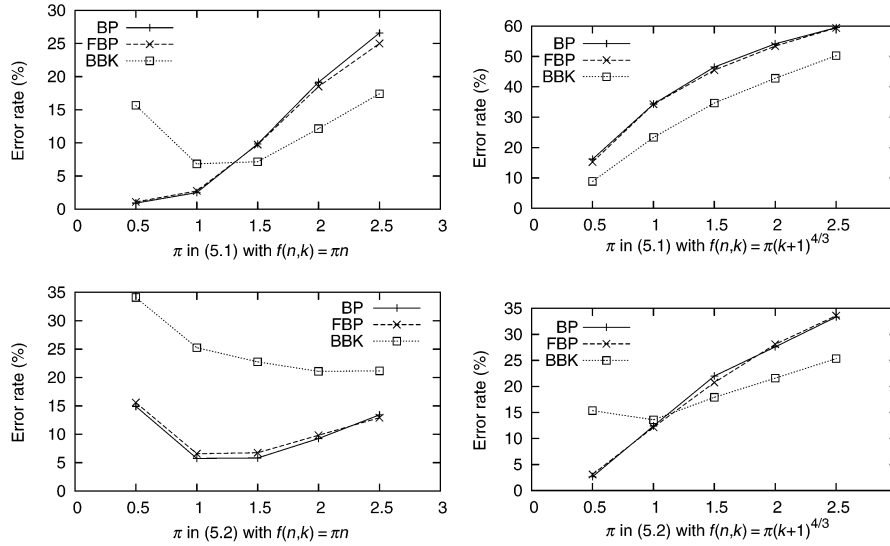

FIG. 5. Error rates using test set B2 of 1200 indefinite matrices of sizes up to $n = 1000$.

The stopping criteria suggested by Higham (1990, p. 183) for rank estimation of positive semidefinite matrices by the Cholesky factorization are in the same form as (5.1) and (5.2) with $f(n, k) = n$.

Forcing all of the nonzero eigenvalues to be positive, we also experimented with rank estimation of symmetric positive semidefinite matrices, which were generated in the same way as the symmetric indefinite matrices but with $t = 0$ (i.e., no negative eigenvalues). Using the same parameters n, r and σ in Table 3, we obtained four sets of matrices $\mathrm{A1}^+$, $\mathrm{A2}^+$, $\mathrm{B1}^+$ and $\mathrm{B2}^+$ corresponding to the sets A1, A2, B1 and B2.

The results of the experiments on these semidefinite matrices were good, even slightly better than the results on the indefinite ones. A difference is that $f(n, k) = \pi n$ is a better choice than $f(n, k) = \pi(k+1)^{4/3}$ for matrices in the sets $\mathrm{A1}^+$ and $\mathrm{B1}^+$ (i.e., with clear ranks). For example, using the stopping criterion (5.2) and complete pivoting, we obtained all estimated ranks accurately with $f(n, k) = \pi n$ for all $\pi = 2, 4, \ldots, 20$, but errors occurred when we set $f(n, k) = \pi(k + 1)^{4/3}$ with $\pi \geqslant 10$. This more or less confirms that the stopping criteria suggested by Higham (1990, p. 183) is a good choice for symmetric positive semidefinite matrices.

Overall, we recommend the fast Bunch–Parlett pivoting strategy and the stopping criterion (5.2) for their efficiency without losing significant accuracy of rank/inertia estimation. The optimal parameters for $f(n, k)$ in (5.2) depend on several factors. *A priori* information about the matrix, such as the ambiguity of the rank, the size of $\|W\|_F$, the growth factor and the distribution of nonzero eigenvalues, may help to adjust the stopping criterion.

6. Concluding remarks

We have studied the stability analysis of block LDL^{T} factorization for symmetric matrices. Our concluding remarks are listed below.

1. A new componentwise backward error analysis of block LDL^{T} factorization for symmetric indefinite matrices is given. It is based on the block LDL^{T} factorization in inner product form,

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

while previous literature uses the outer product formulation. By bridging the gap between the inner and outer product formulations in floating-point computation, our results remain valid for the block LDL^{T} factorization in outer product form.

2. Theorem 3.6 gives a sufficient condition under which a block LDL^{T} factorization of a symmetric matrix is guaranteed to preserve the inertia. The corresponding result for symmetric tridiagonal matrices was given in Corollary 3.7.
3. The block LDL^{T} factorization with Bunch–Parlett, fast Bunch–Parlett or bounded Bunch–Kaufman pivoting (in exact arithmetic) is inertia revealing. The numerical stability of block LDL^{T} factorization in the case without full rank is analysed. The result is given in Theorem 4.4, which supports the reliability of rank estimation and inertia estimation by block LDL^{T} factorization.
4. We have carried out several experiments of rank estimation and inertia estimation by block LDL^{T} factorization. Based on our analysis and numerical experience, we suggest the stopping criteria (5.1) and (5.2) with $f(n,k) = \pi n^p (k+1)^q$, $1 \leqslant p+q \leqslant 3/2$ and $\pi > 0$ modest. The optimal π, p and q depend on the characteristics of the symmetric matrix. Both stopping criteria, with appropriate parameters, worked well with the Bunch–Parlett and fast Bunch–Parlett algorithms when the rank was not ambiguous. We recommend (5.2) for its low cost, and the fast Bunch–Parlett pivoting strategy for its efficiency without losing much accuracy.

Acknowledgements

Part of this work is contained in Fang (2006), the author's doctoral thesis. The Department of Computer Science and Engineering, University of Minnesota, provided computer resources for experiments. Special thanks to Dianne P. O'Leary for constructive discussions at early stages of this work and her help in editing an initial manuscript. Thanks also go to Howard Elman for his careful reading of Fang (2006, Chapter 3), to Nick Higham for his valuable comments, encouragement and providing the Wilkinson (1968) reference and to Che-Rung Lee for helpful discussions on eigenvalues. It is also a pleasure to thank the Associate Editor, Iain S. Duff, for his help in improving the presentation and a referee for the inspiring comments.

Funding

National Science Foundation (CCR 02-04084).

References

Anderson, E., Bai, Z., Bischof, C., Blackford, S., Demmel, J., Dongarra, J., Croz, J. D., Greenbaum, A., Hammarling, S., McKenney, A. & Sorensen, D. (1999) *LAPACK Users' Guide*, 3rd edn. Philadelphia, PA: SIAM.

Ashcraft, C., Grimes, R. G. & Lewis, J. G. (1998) Accurate symmetric indefinite linear equation solvers. *SIAM J. Matrix Anal. Appl.*, **20**, 513–561.

Bunch, J. R. (1971) Analysis of the diagonal pivoting method. *SIAM J. Numer. Anal.*, **8**, 656–680.

Bunch, J. R. (1974) Partial pivoting strategies for symmetric matrices. *SIAM J. Numer. Anal.*, **11**, 521–528.

Bunch, J. R. & Kaufman, L. (1977) Some stable methods for calculating inertia and solving symmetric linear systems. *Math. Comp.*, **31**, 163–179.

Bunch, J. R. & Marcia, R. F. (2005) A pivoting strategy for symmetric tridiagonal matrices. *Numer. Linear Algebra Appl.*, **12**, 911–922.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

BUNCH, J. R. & MARCIA, R. F. (2006) A simplified pivoting strategy for symmetric tridiagonal matrices. *Numer. Linear Algebra Appl.*, **13**, 865–867.

BUNCH, J. R. & PARLETT, B. N. (1971) Direct methods for solving symmetric indefinite systems of linear equations. *SIAM J. Numer. Anal.*, **8**, 639–655.

DONGARRA, J. J., BUNCH, J. R., MOLER, G. B. & STEWART, G. W. (1979) *LINPACK Users' Guide*. Philadelphia, PA: SIAM.

DONGARRA, J. J., DUFF, I. S., SORENSEN, D. C. & VAN DER VORST, H. A. (1991) *Solving Linear Systems on Vector and Shared Memory Computers*. Philadelphia, PA: SIAM.

DUFF, I. S. & REID, J. K. (1982) MA27—a set of Fortran subroutines for solving sparse symmetric sets of linear equations. *Technical Report AERE R10533*. London: AERE Harwell, HMSO.

DUFF, I. S. & REID, J. K. (1983) The multifrontal solution of indefinite sparse symmetric linear systems. *ACM Trans. Math. Softw.*, **9**, 302–325.

FANG, H. (2006) Matrix factorizations, triadic matrices, and modified Cholesky factorizations for optimization. *Ph.D. Thesis*, University of Maryland, MD.

FANG, H. & O'LEARY, D. P. (2006) Stable factorizations of symmetric tridiagonal and triadic matrices. *SIAM J. Matrix Anal. Appl.*, **28**, 576–595.

GOLUB, G. H. & VAN LOAN, C. F. (1996) *Matrix Computations*, 3rd edn. Baltimore, MD: Johns Hopkins University Press.

HIGHAM, N. J. (1990) Analysis of the Cholesky decomposition of a semi-definite matrix. *Reliable Numerical Computation* (M. G. Cox & S. J. Hammarling eds). Oxford: Oxford University Press, pp. 161–185.

HIGHAM, N. J. (1997) Stability of the diagonal pivoting method with partial pivoting. *SIAM J. Matrix Anal. Appl.*, **18**, 52–65.

HIGHAM, N. J. (1999) Stability of block LDL^T factorization of a symmetric tridiagonal matrix. *Linear Algebra Appl.*, **287**, 181–189.

HIGHAM, N. J. (2002) *Accuracy and Stability of Numerical Algorithms*, 2nd edn. Philadelphia, PA: SIAM.

LANCZOS, C. (1950) An iteration method for the solution of the eigenvalue problem of linear differential and integral operators. *J. Res. Natl. Bur. Stand.*, **45**, 255–282.

LANCZOS, C. (1952) Solution of systems of linear equations by minimized iterations. *J. Res. Natl. Bur. Stand.*, **49**, 33–53.

SLAPNIČAR, I. (1998) Componentwise analysis of direct factorization of real symmetric and Hermitian matrices. *Linear Algebra Appl.*, **272**, 227–275.

STEWART, G. W. (1980) The efficient generation of random orthogonal matrices with an application to condition estimation. *SIAM J. Numer. Anal.*, **17**, 403–409.

STEWART, G. W. & SUN, J.-G. (1990) *Matrix Perturbation Theory*. Boston, MA: Academic Press.

WILKINSON, J. H. (1968) A priori error analysis of algebraic processes. *Proceedings of International Congress of Mathematicians*, Moscow 1966 (I. G. Petrovsky ed.). Moscow: Mir, pp. 629–640.

Appendix A. Pivoting methods for general symmetric matrices

This appendix reviews the three types of pivoting algorithms of block LDL^{T} factorization for general symmetric matrices, namely complete pivoting, partial pivoting and rook pivoting.

The current Schur complement is denoted by A, whose (i, j) entry is denoted by a_{ij}. A pivoting method chooses a 1×1 pivot or a 2×2 block pivot from A. The rows and columns are then interchanged, if it is necessary, such that the chosen pivot or block pivot is at the top left corner. Then A is decomposed as (1.1). The decomposition is repeated with the same pivoting method on the next Schur complement until the block LDL^{T} factorization is done.

By Sylvester's law of inertia (Golub & Van Loan, 1996, Theorem 8.1.17), a symmetric matrix has the same inertia as the block diagonal matrix in its block LDL^{T} factorization in exact arithmetic. Therefore,

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

whether a pivoting strategy guarantees an inertia-revealing block LDL^{T} factorization depends on whether it guarantees the completion of factorization (i.e., without leaving a nonzero Schur complement unprocessed). As this appendix will show, both complete pivoting and rook pivoting, but not partial pivoting, guarantee that the block LDL^{T} factorization preserves the inertia.

A.1 *Complete pivoting*

Bunch & Parlett (1971) gave a stable pivoting method presented in Algorithm A.1.

ALGORITHM A.1 Bunch–Parlett pivot selection

$\alpha := (1+\sqrt{17})/8 \approx 0.64$
$a_{kk} :=$ the largest magnitude diagonal element
$a_{ij} :=$ the largest magnitude off-diagonal element ($i < j$)
if $|a_{kk}| \geqslant \alpha |a_{ij}|$ **then**
  Use a_{kk} as a 1×1 pivot ▷ Here if $a_{kk} = 0$, the factorization is done
else
  Use $\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$ as a 2×2 block pivot
end if

The Bunch–Parlett pivoting is most stable. However, at each pivot selection it finds the largest diagonal element and the largest off-diagonal element in magnitude. Therefore factoring a nonsingular symmetric matrix of size $n \times n$ needs between $n^3/12$ and $n^3/6$ comparisons. The cost is of the same order as computing the block LDL^{T} factorization, which takes $n^3/6$ multiplications and $n^3/6$ additions (subtractions).

A.2 *Partial pivoting*

Bunch & Kaufman (1977) devised an efficient pivoting strategy that is presented in Algorithm A.2.

ALGORITHM A.2 Bunch–Kaufman pivot selection

$\alpha := (1+\sqrt{17})/8 \approx 0.64$
$i := 1$ ▷ A variant chooses i such that $|a_{ii}| \geqslant |a_{kk}|$ for all k
Find the index $j \neq i$ such that $|a_{ji}| = \max\{|a_{ki}| : k \neq i\} =: \lambda$
if $|a_{ii}| \geqslant \alpha\lambda$ **then**
  Use a_{ii} as a 1×1 pivot ▷ (*)
else
  $\sigma := \max\{|a_{kj}| : k \neq j\}$ ▷ (†)
  if $|a_{ii}|\sigma \geqslant \alpha\lambda^2$ **then**
    Use a_{ii} as a 1×1 pivot
  else if $|a_{jj}| \geqslant \alpha\sigma$ **then** ▷ (**)
    Use a_{jj} as a 1×1 pivot
  else
    Use $\begin{bmatrix} a_{ii} & a_{ij} \\ a_{ji} & a_{jj} \end{bmatrix}$ as a 2×2 pivot
  end if
end if

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

The Bunch–Kaufman pivoting is not rank revealing because a zero pivot may be chosen while there are still nonzero elements in a Schur complement. On the other hand, it is most economic since at most two columns are visited in each pivot selection, and, overall, only $\mathcal{O}(n^2)$ comparisons are required. To improve the stability a variant, with extra comparisons of diagonal elements, begins with the largest magnitude diagonal pivot rather than a_{11}. For a symmetric positive semidefinite matrix whose maximum magnitude element is on the diagonal, this variant guarantees a 1×1 pivot in (*), and the block LDL^{T} factorization reduces to the Cholesky factorization with complete pivoting.[2] Furthermore, this variant does not require the **else-if** check in (**) because at this point

$$|a_{jj}| \leqslant |a_{ii}| < \alpha\lambda \leqslant \alpha\sigma.$$

These properties were observed by Bunch & Kaufman (1977).

Sorensen and Van Loan (Dongarra *et al.*, 1991, Section 5.3.2) proposed another variant by rewriting (†) in Algorithm A.2 as $\sigma := \max\{|a_{kj}| : \text{for all } k\}$. This small change needs only one more comparison to include the diagonal element at each pivot selection. It has a nice property that, for a symmetric positive definite matrix, the chosen pivot is always the first element a_{11}. As a result, the block LDL^{T} factorization reduces to the Cholesky factorization without pivoting.

A.3 *Rook pivoting*

Despite the efficiency, the Bunch–Kaufman pivoting results in the unit lower triangular matrix L unbounded, which has its practical effects. The Bunch–Parlett pivoting bounds L but it is expensive. The cost for a bounded L can be reduced by a rook searching strategy that is presented in Algorithm A.3 (Duff & Reid, 1982, 1983; Ashcraft *et al.*, 1998).

ALGORITHM A.3 Pivot selection by rook pivoting

```
i := 1 for the bounded Bunch–Kaufman pivoting
i := index of the maximum diagonal element for the fast Bunch–Parlett pivoting
α := (1 + √17)/8 ≈ 0.64
Find the index j ≠ i such that |a_ji| = max{|a_pi| : p ≠ i}
if |a_ii| ⩾ α|a_ji| and a_ii ≠ 0 then
    Use a_ii as a 1 × 1 pivot
else
    repeat
        Find the index k ≠ j such that |a_kj| = max{|a_pj| : p ≠ j}        ▷ (*)
        if |a_jj| ⩾ α|a_kj| and a_jj ≠ 0 then                             ▷ (†)
            Use a_jj as a 1 × 1 pivot
        else if |a_ij| = |a_kj| then
            Use [a_ii  a_ij; a_ji  a_jj] as a 2 × 2 pivot
        else
            Set i := j and j := k
        end if
    until a pivot is chosen, or all columns have been visited
end if
```

[2] In the appendices the Cholesky factorization refers to the $\bar{L}\bar{D}\bar{L}^{\mathrm{T}}$ form, where $\bar{L}$ is unit lower triangular and $\bar{D}$ is diagonal.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

In Algorithm A.3, if the starting element a_{ii} is a_{11}, then it is called the bounded Bunch–Kaufman pivoting. If the starting element a_{ii} is the largest magnitude diagonal element, then it is called the fast Bunch–Parlett pivoting. In the latter case the **if** check in (†) is not required because here

$$|a_{jj}| \leqslant |a_{ii}| < \alpha \max\{|a_{pi} : p \neq i\} \leqslant \alpha |a_{kj}|.$$

As a result, only a 2×2 pivot can be chosen in the loop. See Ashcraft *et al.* (1998) and Duff & Reid (1982, 1983) for more information.

The block LDL^{T} factorization with rook pivoting, as well as that with the Bunch–Parlett pivoting, is inertia revealing. Note, however, that, when the jth column of A in (*) is zero, one needs to avoid duplicate column visits by carefully choosing k.

Appendix B. Pivoting methods for symmetric tridiagonal matrices

This appendix reviews Bunch's method (Bunch, 1974, Section 4) and the Bunch–Marcia pivoting (Bunch & Marcia, 2005, 2006) for block LDL^{T} factorization of a symmetric tridiagonal matrix.

We follow the notation in Appendix A. For simplicity, we assume that the given symmetric tridiagonal matrix is irreducible, or, equivalently, all subdiagonal elements are nonzero. (The treatment of the pivoting algorithms for reducible cases is straightforward.) As this appendix will show, both pivoting strategies choose either a 1×1 pivot a_{11} or a 2×2 block pivot $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ at each pivot selection. As a result, no interchanges of rows and columns are done during the factorization. The decomposition (1.1) only changes the value of the first diagonal entry in the next Schur complement, which therefore remains irreducible. By induction, all of the Schur complements are irreducible.

B.1 *Bunch's method*

Bunch (1974, Section 4) gave a stable pivoting method that is presented in Algorithm A.4.

ALGORITHM A.4 Bunch's pivot selection

$\alpha := (\sqrt{5} - 1)/2 \approx 0.618$
$\sigma :=$ the largest magnitude of elements in the initial given matrix, computed only once ▷ (*)
if $|a_{11}|\sigma \geqslant \alpha |a_{21}|^2$ **then**
  Use a_{11} as a 1×1 pivot
else
  Use $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ as a 2×2 block pivot
end if

In the original formulation of Bunch (1974, Section 4) the first diagonal entry of the given matrix is excluded in (*) while computing σ. Higham (1999, p. 183) proposed a modification to include it to increase the chance that a 1×1 will be chosen without affecting his stability analysis. In either case the resulting block LDL^{T} factorization of a symmetric positive definite tridiagonal matrix reduces to the Cholesky factorization without pivoting.

B.2 *The Bunch–Marcia pivoting*

As shown in (*) in Algorithm A.4, Bunch's pivoting method requires the whole symmetric tridiagonal matrix to be known *a priori*. On the other hand, in some applications, such as applying the Lanczos

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018

method to solve a symmetric indefinite linear system (Lanczos, 1952), it is desired to form the block LDL^{T} factorization while the symmetric tridiagonal matrix is formed. Therefore Bunch & Marcia (2005, 2006) devised a pivoting strategy that needs only the elements in the top left 3×3 block of a Schur complement to determine a pivot, as presented in Algorithm A.5.

ALGORITHM A.5 Bunch–Marcia pivot selection

$\alpha := (\sqrt{5} - 1)/2 \approx 0.618$
$\varDelta := a_{11}a_{22} - a_{21}^2$
if $|\varDelta| \leqslant \alpha |a_{11}a_{32}|$ or $|a_{21}\varDelta| \leqslant \alpha |a_{11}^2 a_{32}|$ or $\underline{|a_{11}a_{22}| \geqslant \alpha a_{21}^2}$ **then** $\triangleright$ (*)
  Use a_{11} as a 1×1 pivot
else
  Use $\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ as a 2×2 block pivot
end if

Note that the original formulation of Bunch & Marcia (2005) did not have the underlined condition $|a_{11}a_{22}| \geqslant \alpha a_{21}^2$ in (*), and for stability reasons the 2×2 linear systems are solved in two different ways, that is, if $|a_{11}a_{22}| \geqslant \alpha a_{21}^2$, then the 2×2 system is solved by the LDL^{T} factorization, otherwise, it is solved by the explicit inverse with scaling (3.2). The underlined condition was later included in the simplified strategy (Bunch & Marcia, 2006), which increases the chance that a 1×1 pivot will be chosen, and has a pleasing effect that, for a symmetric positive definite and tridiagonal matrix, the block LDL^{T} factorization reduces to the Cholesky factorization without pivoting. In addition, this simplified strategy needs only the explicit inverse with scaling (3.2), or GEPP, to solve 2×2 systems.

Downloaded from https://academic.oup.com/imajna/article-abstract/31/2/528/666435
by UCLA Science & Engineering Lib user
on 06 July 2018